「p- Group.

「Thm. Let G be a Group of order p^n for some prime p, $n \in \mathbb{N}$.
Then, G has a Normal Series $\{P_i\}_{i=0}^{n}$ Such that

$$1 = P_0 \triangleleft P_1 \triangleleft \cdots \triangleleft P_{n-1} \triangleleft P_n = G. \quad (|P_i| = p^i)$$

Proof. First, $Z(G) \neq 1$ because The Class Equation: if $Z(G) = 1$,

$$p^n = |G| = |Z(G)| + \sum |G : C_G(g_i)| = 1 + \sum p^{r_i} \quad (1 \leq r_i \leq n-1)$$

This implies p divides 1, Contradiction.
Now, Using induction:
For $n=2$,
$|G/Z(G)| = p$ or 1, thus it must be cyclic, this implies G is Abelian.
By Cauchy's Thm, G has a subgroup of order p, thus Claim is satisfied.
For $n \geq 2$, suppose the claim is true.
Let $|G| = p^{n+1}$. Since $1 \subsetneq Z(G)$ and Cauchy Thm,
there exists a Normal Subgroup $N \trianglelefteq G$ in $Z(G)$ with order p.
Now, G/N is a Group of order p^n, By hypothesis, G/N has a Normal Series:

$$1 = N/N \triangleleft N_1/N \triangleleft \cdots \triangleleft N_{n-1}/N \triangleleft N_n/N = G/N.$$

The fourth Isomorphism gives $N_{n-1} \trianglelefteq G$ and $|N_{n-1}| = p^n$, the claim is proved.

「Lemma. $G/Z(G)$ is cyclic $\Leftrightarrow$ G is Abelian.

Proof. Suppose $G/Z(G)$ is cyclic. Let $xZ(G)$ be a generator of $G/Z(G)$.
Let $a, b \in G$ be given. Then, for some $m, n \in \mathbb{Z}$,

$$aZ(G) = x^m Z(G), \quad bZ(G) = x^n Z(G).$$

Now, $ax^{-m}, bx^{-n} \in Z(G)$.

$$ab = ax^{-m}x^m bx^{-n}x^n = x^{m+n} \cdot bx^{-n} ax^{-m} = ba.$$